University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II

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Second Midterm Exam – Version A

October 30, 2018

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 80 minutes to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

(g) Cellular phones, unauthorized electronic devices or course notes (unless an open-book exam) are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you have ensured that you are complying with the above statement.

Signature _____

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	5	5	4	4	4	2	24
Grade							

1. Consider the vectors $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 3 \\ -5 \\ -2 \\ 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ -2 \\ -1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 10 \\ -18 \\ -6 \\ 6 \end{bmatrix}$.

(a) **[4 points]** Find the general solution of the vector equation

$$a\mathbf{u} + b\mathbf{v} + c\mathbf{w} + d\mathbf{x} = \mathbf{0}$$

for variables a, b, c, d. As usual, the "$\mathbf{0}$" on the right hand side denotes the zero vector.

Solution: We compute the REF of the augmented matrix:

$$[\mathbf{u}\,\mathbf{v}\,\mathbf{w}\,\mathbf{x}|\mathbf{0}] = \left[\begin{array}{cccc|c} 1 & 3 & 1 & 10 & 0 \\ -1 & -5 & 1 & -18 & 0 \\ -1 & -2 & -2 & -6 & 0 \\ 1 & 1 & -1 & 6 & 0 \end{array}\right] \xrightarrow[R_4 \to R_4 - R_1]{\substack{R_2 \to R_2 + R_1 \\ R_3 \to R_3 + R_1}} \left[\begin{array}{cccc|c} 1 & 3 & 1 & 10 & 0 \\ 0 & -2 & 2 & -8 & 0 \\ 0 & 1 & -1 & 4 & 0 \\ 0 & -2 & -2 & -4 & 0 \end{array}\right]$$

$$\xrightarrow[R_4 \to \frac{1}{-2}R_4]{R_2 \to \frac{1}{-2}R_2} \left[\begin{array}{cccc|c} 1 & 3 & 1 & 10 & 0 \\ 0 & 1 & -1 & 4 & 0 \\ 0 & 1 & -1 & 4 & 0 \\ 0 & 1 & 1 & 2 & 0 \end{array}\right] \xrightarrow[R_4 \to R_4 - R_2]{R_3 \to R_3 - R_2} \left[\begin{array}{cccc|c} 1 & 3 & 1 & 10 & 0 \\ 0 & 1 & -1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \left[\begin{array}{cccc|c} 1 & 3 & 1 & 10 & 0 \\ 0 & 1 & -1 & 4 & 0 \\ 0 & 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_3 \to \frac{1}{2}R_3} \left[\begin{array}{cccc|c} 1 & 3 & 1 & 10 & 0 \\ 0 & 1 & -1 & 4 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 + R_3} \left[\begin{array}{cccc|c} 1 & 3 & 0 & 11 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to R_1 - 3R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

The last matrix is the augmented matrix of the linear system

$$\begin{cases} a + 2d = 0 \\ b + 3d = 0 \\ c - d = 0 \end{cases}$$

We obtain

$$\begin{cases} a = -2d \\ b = -3d \\ c = d \\ d : \text{free.} \end{cases}$$

(b) **[1 point]** Is the set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}\}$ linearly dependent?

- If it is linearly independent, then explain why;
- If it is linearly dependent, then find an explicit linear dependence relation for these vectors.

Solution: From (a) we see that the vector equation has non-trivial solution. Thus the set is linearly **dependent**.

Take for example $d = 1$, then $a = -2$, $b = -3$, $c = 1$. Thus

$$-2\mathbf{u} - 3\mathbf{v} + \mathbf{w} + \mathbf{x} = \mathbf{0}.$$

2. Let $B = \begin{bmatrix} 1 & -3 \\ 2 & -7 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 4 \\ -5 & -6 \end{bmatrix}$.

(a) **[3 points]** Find the 2×2 matrix A such that $A = (C^T + I_2)B^{-1}$, where I_2 is the 2×2 identity matrix.

Solution:

$$C^T + I_2 = \begin{bmatrix} 0 & -5 \\ 4 & -6 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -5 \\ 4 & -5 \end{bmatrix},$$

$$B^{-1} = \frac{1}{-1}\begin{bmatrix} -7 & 3 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -3 \\ 2 & -1 \end{bmatrix},$$

$$A = \begin{bmatrix} 1 & -5 \\ 4 & -5 \end{bmatrix}\begin{bmatrix} 7 & -3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 18 & -7 \end{bmatrix}.$$

(b) **[2 points]** Calculate B^{-2}.

Solution:

$$B^{-2} = B^{-1}B^{-1} = \begin{bmatrix} 7 & -3 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 7 & -3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 43 & -18 \\ 12 & -5 \end{bmatrix}.$$

3. [**4 points**] For each of the following statements, indicate if it is true (T) or false (F). You will receive 1 point for each correct answer, 0 points for leaving each answer space blank, and -0.5 points for each incorrect answer (but you will not receive a negative total score on this question).

_____The set $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ is linearly independent.

_____For any given 2×2 matrices A, B and C such that $A \neq 0_{2\times 2}$, if $AB = AC$ then we necessarily have $B = C$.

_____If the columns of a 4×4 matrix A are linearly independent, then the linear system $A\mathbf{x} = \mathbf{0}$ has no solutions other than the trivial solution.

_____For any two 3×3 matrices A and B, we have $(A^T B^T)^T = AB$.

Solution: T F T F.

4. **[4 points]** Find the inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 2 & 7 & 6 \\ 3 & 9 & 10 \end{bmatrix}$.

Solution:

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 3 & 9 & 10 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 3R_3} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 10 & 0 & -3 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to R_1 - 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 16 & -3 & -3 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array}\right].$$

Hence, $A^{-1} = \begin{bmatrix} 16 & -3 & -3 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$.

5. Consider an economy with two sectors: Agriculture and Service. For each unit of output, the Agriculture sector requires $\frac{1}{3}$ units from Agriculture and $\frac{1}{2}$ units from Service. For each unit of output, the Service sector requires $\frac{1}{6}$ units from Agriculture and $\frac{3}{4}$ units from Service.

(a) **[1 point]** Write the consumption matrix C for this economy.

(b) **[1 point]** Find $I - C$ and $(I - C)^{-1}$.

(c) **[2 points]** Find the production levels that are needed to satisfy the final demand of 20 units from the Agriculture sector and 10 units from the Service sector.

Solution: (a) The consumption matrix of this economy is:

$$C = \begin{bmatrix} \frac{1}{3} & \frac{1}{6} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix}$$

(b)

$$I - C = \begin{bmatrix} 1 - \frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{2} & 1 - \frac{3}{4} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

$$(I - C)^{-1} = \frac{1}{\frac{1}{12}} \begin{bmatrix} \frac{1}{4} & \frac{1}{6} \\ \frac{1}{2} & \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 6 & 8 \end{bmatrix}$$

(c) $\mathbf{d} = \begin{bmatrix} 20 \\ 10 \end{bmatrix}$.

$$\mathbf{x} = (I - C)^{-1}\mathbf{d} = \begin{bmatrix} 3 & 2 \\ 6 & 8 \end{bmatrix} \begin{bmatrix} 20 \\ 10 \end{bmatrix} = \begin{bmatrix} 80 \\ 200 \end{bmatrix}$$

6. [**2 points**] Suppose that A, B, and C are 2×2 invertible matrices. Solve the matrix equation

$$A(B^{-1} + X^T)C = CA$$

for X (i.e., express X in terms of A, B, and C).

Solution:

$$A(B^{-1} + X^T)CC^{-1} = CAC^{-1}.$$
$$A(B^{-1} + X^T) = CAC^{-1}.$$
$$A^{-1}A(B^{-1} + X^T) = A^{-1}CAC^{-1}.$$
$$B^{-1} + X^T = A^{-1}CAC^{-1}.$$
$$X^T = A^{-1}CAC^{-1} - B^{-1}.$$
$$X = (A^{-1}CAC^{-1} - B^{-1})^T.$$